

BIO-DATA

Dr. Debashis Chatterjee

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Last updated: 10 June 2026

1. Name and full correspondence address

Name: Dr. Debashis Chatterjee

Designation: Assistant Professor in Statistics; Adjunct Fellow, S. N. Bose National Centre for Basic Sciences, Kolkata.

Official Correspondence Address: Department of Statistics, Siksha-Bhavana (Institute of Science), Visva-Bharati University, P.O. Santiniketan, West Bengal, India, Pin 731235.

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2. Email(s) and contact number(s)

Official email: debashis.chatterjee@visva-bharati.ac.in

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3. Institution

Primary Institution: Department of Statistics, Siksha-Bhavana (Institute of Science), Visva-Bharati University, Santiniketan, West Bengal, India.

Adjunct Affiliation: Adjunct Fellow, S. N. Bose National Centre for Basic Sciences, Kolkata.

4. Date of Birth: 06 February 1991

5. Gender: Male

6. Category: General

7. Differently abled: No

8. Academic Qualification (Undergraduate onwards)

S.No.	Degree	Year	Subject	University/Institution	% of marks
1	B.Stat. (Honours)	2013	Statistics	Indian Statistical Institute	First Division
2	M.Stat.	2015	Mathematical Statistics and Probability	Indian Statistical Institute	First Division
3	Ph.D.	2022; degree received on 24 May 2022	Statistics	Indian Statistical Institute	Not applicable

9. Ph.D. thesis Details

Thesis title: *A Brief Treatise on Bayesian Inverse Regression.*

Guide/Supervisor: Prof. Saurabh Bhattacharya, Interdisciplinary Statistical Research Unit, Indian Statistical Institute.

Institute: Indian Statistical Institute.

Thesis submitted: December 2020.

Year of award/degree received: 2022; exact date of degree received: **24 May 2022.**

Repository link: <http://library.isical.ac.in:8080/jspui/handle/10263/7094>.

Long version: https://www.researchgate.net/publication/360456798_A_BRIEF_TREATISE_ON_BAYESIAN_INVERSE_REGRESSION.

10. Work experience (in chronological order)

S.No.	Position held	Name of the Institute	From	To	Pay Scale
1	Research Scholar/Ph.D. Scholar	Indian Statistical Institute	Dec. 2015	Dec. 2020	As per institute rules
2	Assistant Professor in Statistics	Department of Statistics, Siksha-Bhavana, Visva-Bharati University	02 July 2021	Present	Academic Level 10, 7th CPC

11. Professional Recognition/Award/Prize/Certificate/Fellowship received

S.No.	Name of Award/Recognition	Awarding Agency	Year
1	Professor Aditya Chattopadhyay Young Scientist Award (PACM YSA 2025), announcement link	Calcutta Statistical Association	2026
2	Best Teacher Honor based on student feedback	Siksha-Bhavana Library, Visva-Bharati	2024
3	Best Paper Presentation Award, National Seminar on Application of Statistics on Natural Science	IUCAA Centre for Astronomy Research and Development, St. Xavier's College (Autonomous), Kolkata	2019
4	Certificate of Merit, Indian Standard Examination in Astronomy, top 1% statewide; also qualified Regional Mathematical Olympiad	National Board for Higher Mathematics/DAE-related Olympiad system	2009
5	Featured in news article related to published research on bird migration	The Telegraph India	2024
6	Featured in news article related to published research on volcano-earthquake association	Bartaman, India	2025

Research Guidance: Existing Ph.D. Scholars

The following Ph.D. scholars are ongoing under my supervision/co-supervision. Thesis titles/areas are stated in their current/provisional academic form wherever final submission is pending.

S.No.	Name of Scholar	Institution / Department	Broad Thesis Area / Provisional Title	Role	Current Status
1	Mr. Sagnik Acharyya	Department of Statistics, Visva-Bharati University	On Bayesian Statistical Epidemiology	Supervisor	Thesis work at the final stage; on the verge of defence / submission process
2	Mr. Subhrajit Saha	Department of Statistics, Visva-Bharati University	Directional and Circular Statistics, Statistical Modelling, Environmental / Scientific Data Analysis, and AI/ML-assisted Statistical Applications	Supervisor	Ongoing, Thesis work at the final stage;

Research Proposals Submitted / Institutionally Endorsed / Under Consideration

The following proposals are listed as submitted, institutionally endorsed, or under consideration. They are not claimed as sanctioned/funded projects unless an official sanction order is received. The starred item indicates a proposal presently at an advanced/final stage of consideration.

S.No	Scheme / Agency	Title of Proposal	Role	Collaborators / Team	Status
1*	ANRF AI for Science & Engineering / AI Weather and Climate Modelling	Spherical & Bayesian Inverse Intelligence for Indian Weather, Climate, and Geological Hazards	Lead PI	Prof. Amlan Banerjee, Geological Studies Unit, Indian Statistical Institute, Kolkata	Final-stage evaluation / under final consideration
2	ANRF ARG–MATRICS	Stochastic Directional Operations Research on Manifolds: Distributional Theory, Bayesian Extension, Geodesic Risk and Applications in Astronomy and Bioscience	PI	—	Submitted under ANRF ARG–MATRICS
3	ANRF Advanced Research Grant / ARG	Higher-Order Measure-Theoretic Differentiation and Singularity Geometry: A Unified Calculus for Irregular Scientific Data, Geometric Measures and India-Relevant Extreme Phenomena	Co-PI	PI: Dr. Subhasis Roy	Departmentally recommended / forwarded for institutional endorsement
4	Visva-Bharati Internally Funded Research Grant for Young Faculties, 2026	AI-Mediated Interpretive Responses to Tagore and Wordsworth across Transcendental, Relational and Environmental Axes	Co-PI	PI: Dr. Surhita Basu; Co-PIs: Dr. Shinjini Basu and Dr. Debashis Chatterjee	Submitted / processed internally at Visva-Bharati

* Proposal with Prof. Amlan Banerjee is at an advanced/final stage of consideration.

12. Publications

Note: Published/accepted journal papers are listed in year-wise descending order. The serial numbers are deliberately reverse chronological: the latest/top items carry larger serial numbers, and the oldest/bottom item carries S.No. 1. Manuscripts under revision/revise-and-resubmit/submitted are placed in a separate subsection.

12A. Published/Accepted journal papers, year-wise descending order

S.No.	Author(s)	Title	Name of Journal	Volume/Page/DOI	Year/Status
33	Saha, S. and Chatterjee, D.*	Mixed Effects and Threshold Regression Modeling of Forest Carbon Dynamics under Disturbance in the FoRTE Experiment	Discover Forests	Accepted; DOI TBA	2026
32	Saha, S. and Chatterjee, D.	Directional statistical modelling of tropical storm tracks and intensities: A novel framework for forecasting and preparedness	Journal of Statistical Research	Accepted	2026
31	Saha, S. and Chatterjee, D.	A novel multivariate generalized circular mixed model for count data: Classical and Bayesian inferences	Austrian Journal of Statistics	Accepted; DOI TBA	2026
30	Banerjee, N., Mazumdar, S. M., Chatterjee, D. and Mazumdar, A.	Delineating wing sexual dimorphism in <i>Culicoides</i> using novel Bayesian and classical geometric morphometrics with notes on allometry	Acta Tropica	278, 108088; DOI: 10.1016/j.actatropica.2026.108088	2026
29	Chatterjee, D. and Ghosh, P.	Oriented Odyssey of the Lunar Scars: Directional Distribution of Lunar Craters	Journal of Space Safety Engineering	DOI: 10.1016/j.jsse.2026.02.009	2026
28	Saha, S. and Chatterjee, D.	Statistical Analysis of Seismicity in the Fiji-Tonga Region: Integrating Classical and Circular Methods	Journal of Statistical Research	59(2), 145–166; DOI: 10.3329/jsr.v59i2.88019	2026
27	Acharyya, S., Rana, S. and Chatterjee, D.	MechGNN-Epi: Mechanistically Constrained Spatiotemporal Graph Learning for Regional Epidemic Forecasting	Model Assisted Statistics and Applications	Accepted; DOI TBA	2026
26	Chatterjee, D., Chandra, N. K. and Maity, S. I.	A novel classical and Bayesian adaptation of the Pólya urn with reinforcement dynamics: Methodologies and applications	Communications in Statistics – Theory and Methods	Accepted; DOI: 10.1080/03610926.2026.2686273	2026

25	Banerjee, M., Chatterjee, D. and Chakraborty, A.	Current trends in COVID-19 severity: A machine learning approach based on recent variants, genomic mutations and demographic information	Brazilian Archives of Biology and Technology	Accepted; DOI TBA	2026
24	Acharyya, S., Saha, S. and Chatterjee, D.	A machine-learning framework for virus early warning from weather and urban sustainability signals	Explora: Environment and Resource	DOI: 10.36922/EER025440076 ; invited paper	2026
23	Chatterjee, D. and Bhattacharya, S.	How ominous is the premonition of future global warming?	Sankhya Series B	DOI: 10.1007/s13571-025-00390-y	2025
22	Chatterjee, D. and Ghosh, P.	Whisperers of whales wander: A directional statistical investigation of whales' migration influenced by geomagnetic, ocean current and celestial cues	Journal for Nature Conservation	127011; DOI: 10.1016/j.jnc.2025.127011	2025
21	Chatterjee, D. and Ghosh, P.	On the directional nature of the fall of celestial objects on the surface of Venus	Planetary and Space Science	106167; DOI: 10.1016/j.pss.2025.106167	2025
20	Chatterjee, D. and Ghosh, P.	Bayesian hierarchical modeling of mucosal immune responses and growth efficiency in young animals: Demonstrating the superiority of data-dependent empirical priors	PLOS ONE	20(6), e0326273; DOI: 10.1371/journal.pone.0326273	2025
19	Banerjee, N., Mazumdar, S. M., Chatterjee, D. and Pal, A.	A novel Bayesian approach based on wing geometric morphometry to discriminate <i>Culicoides</i> species (Diptera: Ceratopogonidae)	Journal of Medical Entomology	DOI: 10.1093/jme/tjaf082	2025
18	Chatterjee, D., Saha, S. and Ghosh, P.	Circular insights for rhythmic health: A Bayesian approach with stochastic diffusion for characterizing human physiological rhythms with applications to arrhythmia detection	PLOS ONE	20(6), e0324741; DOI: 10.1371/journal.pone.0324741	2025
17	Hore, M., Bhattacharyya, S., Chatterjee, D., Saha, S., Biswas, J. K. and Roy, S.	Assessing water quality of River Mahananda, Malda, India using <i>Cirrhinus reba</i> as a bioindicator: oxidative stress biomarkers and histopathological fingerprints	Environmental Monitoring and Assessment	197, 530; DOI: 10.1007/s10661-025-13969-9	2025
16	Chatterjee, D., Ghosh, P., Banerjee, A. and Das, S. S.	Novel empirical models and comparative probabilistic analysis of interconnectedness of volcano eruption and nearby earthquakes of the Earth	PLOS ONE	20(4), e0320210; DOI: 10.1371/journal.pone.0320210	2025
15	Ghosh, P., Ghosh, S. and Chatterjee, D.	A Flexible Spherical Mixture Model for Gamma-Ray Burst Patterns Obtained from BATSE and FERMI Mission	Publications of the Astronomical Society of the Pacific	137(2), 024503; DOI: 10.1088/1538-3873/adb2c1	2025
14	Chatterjee, D. and Ghosh, P.	Redshift-Agnostic Machine Learning Classification: Unveiling Peak Performance in Galaxy, Star, and Quasar Classification using SDSS DR17	Astronomische Nachrichten	e20240057; DOI: 10.1002/asna.20240057	2025
13	Chatterjee, D. and Saha, S.	The philosophy of circular statistical thinking in human cognition, temporal constructs and anthropological studies	Journal of Contemporary Philosophical and Anthropological Studies	3(1); DOI TBA	2025
12	Chatterjee, D. and Ghosh, P.	Mercury's Meteorite Mysteries: A Directional Statistical Guide to Mercury's North Pole, Hidden Hazards and Roadmap to Safe Landing Havens	Publications of the Astronomical Society of the Pacific	136(11), 114509; DOI: 10.1088/1538-3873/ad851b	2024
11	Chatterjee, D., Ghosh, P., Banerjee, A. and Das, S. S.	Optimizing machine learning for water safety: dimensionality reduction and classifier performance in potability prediction	PLOS Water	3(8), e0000259; DOI: 10.1371/journal.pwat.0000259	2024
10	Ghosh, P., Chatterjee, D., Banerjee, A. and Das, S. S.	Do magnetic murmurs guide birds? A directional statistical investigation for influence of Earth's magnetic field on bird navigation	PLOS ONE	19(6), e0304279; DOI: 10.1371/journal.pone.0304279	2024
9	Ghosh, P., Chatterjee, D. and Banerjee, A.	On the directional nature of celestial object's fall on the Earth, Part 1: Distribution of fireball shower, meteor fall and crater on Earth's surface	Monthly Notices of the Royal Astronomical Society	stae1066; DOI: 10.1093/mnras/stae1066	2024
8	Chatterjee, D. and Saha, S.	Novel methodology and optimization of heat pump efficiency through stochastic finite element analysis and circular statistics	Thermal Science and Engineering	7(4), 8795; DOI: 10.24294/tse8795	2024
7	Saha, S. and Chatterjee, D.	A novel circular dynamics in financial networks with cross-correlated volatility and asset movements	Financial Statistical Journal	7(2), 9177; DOI: 10.24294/fsj9177	2024
6	Chatterjee, D.	The impact of algorithmic culture on the construction of self in a globalized world	Journal of Contemporary Philosophical and Anthropological Studies	2(3); DOI: 10.59652/jc-pas.v2i3.199	2024

5	Chatterjee, D., Maitra, T. and Bhattacharya, S.	A short note on almost sure convergence of Bayes factors in the general set-up	The American Statistician	74(1), 17–20; DOI: 10.1080/00031305.2017.1397548	2020
4	Chatterjee, D. and Bhattacharya, S.	A Statistical Perspective on Inverse and Inverse Regression Problems	RASHI	2(1), 67–82; URL available	2017
3	Chatterjee, D.	A Novel Bayes Factor for Inverse Model Selection Problem based on Inverse Reference Distribution	International Journal of Mathematics and Statistics Invention	5(6), 06–09; URL available	2017
2	Sarkar, A., Reiger, R., Chatterjee, D., Patranabis, S., Singh, H. and Mukherjee, P.	Simulation study of a constant time hybrid approach for large scale terrain mapping using satellite stereo imagery	Robotics and Autonomous Systems	79, 132–146; DOI: 10.1016/j.robot.2016.01.004	2016
1	Chatterjee, D.	On consistency issue of the definition of mean free path on the basis of measure theoretic probability and speed of light postulate of relativity	AIJRSTEM	14–116; URL available	2014

12B. Manuscripts under revision / revise-and-resubmit / submitted / communicating

S.No.	Author(s)	Title	Journal/Agency	Status
17	Chatterjee, D.	Decentralized game-theoretic scheduling for electromagnetic follow-up of gravitational-wave events on sky maps	Monthly Notices of the Royal Astronomical Society; Ref. MN-26-0979-P	Revision
16	Acharyya, S. and Chatterjee, D.	How Climate and Connectivity Pattern Epidemic Hotspots of India: A Denominator-Aware Spatio-Temporal Epidemiology Inference of COVID-19 via Binomial GLMMs, GEEs, and Markov Persistence	Journal of the Indian Society for Probability and Statistics, Springer	Major revision
15	Maity, S. I. and Chatterjee, D.	AI-Modulated Pólya Urns (AIM-PU): A Unified Framework for Risk-Sensitive Contextual Bandits, Resource Allocation and Portfolio Design	International Journal of Data Science and Analytics, Springer	Minor revision
14	Chatterjee, D. and Maity, S. I.	Convergence and Non-Convergence in Generalized Pólya Urn Models under Tenable Dynamics: A Theoretical and Computational Exploration with Counterexamples	Calcutta Statistical Association Bulletin, Sage	Revision completed
13	Banerjee, M., Chatterjee, D. and Chakraborty, A.	A Computationally Efficient Framework using Generalized Additive Models for COVID-19: A Short Term Incidence Prediction and Pandemic Dynamics Assessment of India	Sankhya Series B, Springer	Major revision
12	Saha, S. and Chatterjee, D.	A Mixed-Effects and Threshold Regression Approach to Forest Carbon Dynamics under Disturbance: Insights from the FoRTE Experiment	Discover Forests, Springer	Major revision completed
11	Chatterjee, D., Acharyya, S. and Bhattacharya, S.	Joint Monotone Ordinal Learning of Cardiac and Hepatic Iron Overload from Multi-Echo T2* MRI and Clinical Biomarkers: Evidence from the CHMMOTv1 Cohort	PLOS ONE	Under revision
10	Chatterjee, D. and Ghosh, P.	Comparative Analysis of Machine Learning Algorithms for Breast Cancer Classification: SVM Outperforms XGBoost, CNN, RNN, and Others	Current Science	Minor revision
9	Saha, S. and Chatterjee, D.	Spatially Varying Circular-Coefficient Survival Model for Wind Turbine Reliability Analysis	Sankhya Series B	Revise and resubmit
8	Ghosh, P., Chatterjee, D., Banerjee, A., Das, S., Mukherjee, D. and Das, P.	On Spatio-temporal Dependence of Bird's Migration Path Direction on the Change of the Earth's Magnetic Field, Sun and the Moon Positions (European Region Study)	Proceedings of the National Academy of Sciences	Revise and resubmit ongoing
7	Chatterjee, D.	Bayesian Linear Programming under Learned Uncertainty: Posterior Feasibility Guarantees, Scenario Certification, and Applications	Journal of the Royal Statistical Society: Series C	Submitted; arXiv DOI: 10.48550/arXiv.2603.05885
6	Chatterjee, D.	A Hierarchical Bayesian Dynamic Game for Competitive Inventory and Pricing under Incomplete Information: Learning, Credible Risk, and Equilibrium	Journal of Applied Statistics	Submitted; arXiv DOI: 10.48550/arXiv.2603.06072
5	Chatterjee, D.	Bayesian Statistical Genetics Meets Operations Research: A Distributionally Robust Framework for Pathway Discovery and Budgeted Validation	Biometrics	Submitted
4	Chatterjee, D. and Chatterjee, S.	Transport-ReactionNet: A transportation-problem-driven deep learning framework for organic reaction prediction, yield modeling, and condition allocation	Communications in Statistics	Communicating
3	Das, S. and Chatterjee, D.	Distributional genetic association testing via optimal transport geometry on discrete genotype spaces	Philosophical Transactions of the Royal Society	Communicating
2	Chatterjee, D.	Adaptive variance-reduced gradient with momentum (AVRGM): A novel adaptive variance-reduced stochastic gradient algorithm for non-convex optimization	Noted in publication list	Submitted

1	Chatterjee, D.	Advanced trajectory optimization for space missions: A comprehensive mathematical framework	Applied Mathematics and Optimization	Submitted
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12C. Selected arXiv/preprints and other research outputs

S.No.	Author(s)	Title	Repository/Status	Year
11	Chatterjee, D.	Bayesian–AI Fusion for Epidemiological Decision Making: Calibrated Risk, Honest Uncertainty, and Hyperparameter Intelligence	arXiv; DOI: 10.48550/arXiv.2511.11983 ; communicating in Annals of Applied Statistics	2026
10	Chatterjee, D.	Bayesian Conservative Policy Optimization (BCPO): A Novel Uncertainty-Calibrated Offline Reinforcement Learning with Credible Lower Bounds	arXiv; DOI: 10.48550/arXiv.2603.12284	2026
9	Chatterjee, D.	Bayesian Linear Programming under Learned Uncertainty: Posterior Feasibility Guarantees, Scenario Certification, and Applications	arXiv; DOI: 10.48550/arXiv.2603.05885	2026
8	Chatterjee, D.	A Hierarchical Bayesian Dynamic Game for Competitive Inventory and Pricing under Incomplete Information	arXiv; DOI: 10.48550/arXiv.2603.06072	2026
7	Chatterjee, D. and Bhattacharya, S.	How Ominous is the Future Global Warming Premonition?	arXiv:2008.11175; later published in Sankhya Series B	2020
6	Chatterjee, D. and Bhattacharya, S.	A Bayesian Multiple Testing Paradigm for Model Selection in Inverse Regression Problems	arXiv:2007.07847	2020
5	Chatterjee, D. and Bhattacharya, S.	Convergence of Pseudo-Bayes Factors in Forward and Inverse Regression Problems	arXiv:2006.06020	2020
4	Chatterjee, D. and Bhattacharya, S.	Posterior Consistency of Bayesian Inverse Regression and Inverse Reference Distributions	arXiv:2005.00236	2020
3	Chatterjee, D. and Bhattacharya, S.	Posterior Convergence of Nonparametric Binary and Poisson Regression under Possible Misspecifications	arXiv:2005.00234	2020
2	Chatterjee, D. and Bhattacharya, S.	Posterior Convergence of Gaussian and General Stochastic Process Regression under Possible Misspecifications	arXiv:1810.10495	2018
1	Chatterjee, D. and Bhattacharya, S.	A Statistical Perspective on Inverse and Inverse Regression Problems	arXiv:1707.06852; published journal version listed above	2017

13. Detail of patents

S.No.	Patent Title	Name of Applicant(s)	Patent No.	Award Date	Agency/Country	Status
1	Nil	Nil	Nil	Nil	Nil	Nil

14. Books/Reports/Chapters/General articles/software etc.

S.No	Title	Author's Name	Publisher/Repository	Year
6	BPviGM1: Novel Bayesian Procrustes Variance-Based Inferences in Geometric Morphometrics	Chatterjee, D.	Published R package/GitHub package; https://rdrr.io/github/debashischatterjee111/BPviGM1/ ; associated paper DOI: 10.1093/jme/tjaf082	2021–2024
5	Enhancing Outlier Detection in Circular Time Series: A Statistical Framework Leveraging Circular Z-Scores and Dynamic Thresholding	Chatterjee, D. and Saha, S.	Calcutta Statistical Association Bulletin; Eleventh International Triennial Calcutta Symposium	2024
4	Hierarchical Bayesian modeling of heterogeneous circular data with applications in phase analysis	Chatterjee, D. and Saha, S.	Calcutta Statistical Association Bulletin; Eleventh International Triennial Calcutta Symposium	2024
3	A deeper insight into directional skewness and kurtosis in circular statistics	Chatterjee, D. and Saha, S.	Springer Proceedings; RAMSDS 2024, Adamas University	2024

2	Revisit to a Prominent Example of The Curse of Dimensionality: Re-investigating Anomaly Behaviour of Monte Carlo Estimate of π in Higher Dimension	Ghosh, P., Sanyal, S. and Chatterjee, D.	Proceedings of IMBIC Vol. 10, ISBN 978-81-925832-9-7; accepted for Scopus-indexed Springer book series according to CV note	2023
1	Simulation Study of a Constant Time Hybrid Approach for Large Scale Terrain Mapping Using Satellite Stereo Imagery, Chapter 6 in <i>Advances in Robotics and Automatic Control: Reviews</i> , Vol. 1	Sarkar, A., Reiger, R., Chatterjee, D., Patranabis, S., Singh, H. and Mukherjee, P.	International Frequency Sensor Association Publishing, ISBN 978-84-09-02449-0	2018

Ph.D. Thesis Related Other Papers

6. **Chatterjee, Debashis**, and Sourabh Bhattacharya. "Posterior Convergence of Gaussian and General Stochastic Process Regression Under Possible Misspecifications." arXiv:1810.10495 (2018). Available at <https://arxiv.org/pdf/1810.10495>
5. **Chatterjee, Debashis**, and Sourabh Bhattacharya. "Posterior Convergence of Nonparametric Binary and Poisson Regression Under Possible Misspecifications." arXiv:2005.00234 (2020). Available at <https://arxiv.org/pdf/2005.00234>
4. **Chatterjee, Debashis**, and Sourabh Bhattacharya. "Posterior Consistency of Bayesian Inverse Regression and Inverse Reference Distributions." arXiv:2005.00236 (2020). Available at <https://arxiv.org/pdf/2005.00236>
3. **Chatterjee, Debashis**, and Sourabh Bhattacharya. "Convergence of Pseudo-Bayes Factors in Forward and Inverse Regression Problems." arXiv:2006.06020 (2020). Available at <https://arxiv.org/pdf/2006.06020>
2. **Chatterjee, Debashis**, and Sourabh Bhattacharya. "A Bayesian Multiple Testing Paradigm for Model Selection in Inverse Regression Problems." arXiv:2007.07847 (2020). Available at <https://arxiv.org/pdf/2007.07847>
1. **Chatterjee, Debashis**, and Sourabh Bhattacharya. "How Ominous is the Future Global Warming Premonition?." arXiv:2008.11175 (2020). Available at <https://arxiv.org/pdf/2008.11175.pdf> (Now published in *Sankhya*, Series B in 2025)

Selected AI/ML Related Theory /Application Papers: Published/ Accepted

8. **Debashis Chatterjee** and Saran Ishika Maity. AI-Modulated Pólya Urns (AIM-PU): A Unified Framework for Risk-Sensitive Contextual Bandits, Resource Allocation and Portfolio Design. **International Journal of Data Science and Analytics, Springer**. Under minor revision. Research Square DOI: <https://doi.org/10.21203/rs.3.rs-8390479/v1>.
7. Sagnik Acharyya, Subrata Rana and **Debashis Chatterjee***. MechGNN-Epi: Mechanistically Constrained Spatiotemporal Graph Learning for Regional Epidemic Forecasting. **Model Assisted Statistics and Applications, Sage**. Accepted.
6. Meghna Banerjee, **Debashis Chatterjee** and Arindom Chakraborty. Current Trends in COVID-19 Severity: A Machine Learning Approach Based on Recent Variants, Genomic Mutations and Demographic Information. **Brazilian Archives of Biology and Technology**. Accepted, 2026.
5. Sagnik Acharyya, Subhrajit Saha and **Debashis Chatterjee***. A Machine-Learning Framework for Virus Early Warning from Weather and Urban Sustainability Signals. **Explora: Environment and Resource**. Accepted, 2026. DOI: <https://doi.org/10.36922/EERO25440076>.
4. **Debashis Chatterjee** and Prithwish Ghosh. Redshift-Agnostic Machine Learning Classification: Unveiling Peak Performance in Galaxy, Star, and Quasar Classification using SDSS DR17. **Astronomische Nachrichten**. Published, 2025. DOI: <https://doi.org/10.1002/asna.20240057>.

3. **Debashis Chatterjee**, Prithwish Ghosh, Amlan Banerjee and Shiladri Shekhar Das. Optimizing Machine Learning for Water Safety: A Comparative Analysis with Dimensionality Reduction and Classifier Performance in Potability Prediction. **PLOS Water**, 3(8), e0000259. **Published, 2024**. DOI: <https://doi.org/10.1371/journal.pwat.0000259>.
2. Sarkar, A., Reiger, R., **Chatterjee, D.**, Patranabis, S., Singh, H. and Mukherjee, P. Simulation Study of a Constant Time Hybrid Approach for Large Scale Terrain Mapping using Satellite Stereo Imagery. **Robotics and Autonomous Systems**, 79, 132–146. **Published, 2016**. DOI: <https://doi.org/10.1016/j.robot.2016.01.004>.
1. **Debashis Chatterjee**. Bayesian–AI Fusion for Epidemiological Decision Making: Calibrated Risk, Honest Uncertainty and Hyperparameter Intelligence. Communicating in **Annals of Applied Statistics (AOAS)**. arXiv DOI: <https://doi.org/10.48550/arXiv.2511.11983>.

Selected Ongoing papers on AI /ML Core Theory (Published/ Accepted/ Communicating, ArXiv Available)

1. Debashis Chatterjee. Bayesian–AI Fusion for Epidemiological Decision Making: Calibrated Risk, Honest Uncertainty, and Hyperparameter Intelligence. Communicating in Annals of Applied Statistics (AOAS). <https://doi.org/10.48550/arXiv.2511.11983>

Published Software / R Package

Novel R Package Developed: BPviGM1

Title: Bayesian Procrustes Variance-based Inferences in Geometric Morphometrics, Part 1.

- Developed and released an original R package, BPviGM1, for Bayesian Procrustes variance-based inference in geometric morphometrics.
- The package is publicly available through GitHub and indexed in **rdrr**, a comprehensive index of R packages and documentation from CRAN, Bioconductor, GitHub and R-Forge.
- GitHub repository: <https://github.com/debashischatterjee111/BPviGM1>
- rdrr documentation page: <https://rdrr.io/github/debashischatterjee111/BPviGM1/>
- Associated published reference paper: Banerjee, N., Mazumdar, S. M., Chatterjee, D. and Pal, A. (2025). *A novel Bayesian approach based on wing geometric morphometry to discriminate Culicoides species (Diptera: Ceratopogonidae)*. *Journal of Medical Entomology*. DOI: <https://doi.org/10.1093/jme/tjaf082>

Teaching Summary

Semester-wise teaching assignments at Visva-Bharati University, Department of Statistics.

Fall Semester – Visva-Bharati University

August 2021 – December 2021

- BSC-SEM-6 CC-14: Real Analysis.
- MSC-CC-25: Statistical Inference II.
- MSC-21: Measure Theory and Advanced Real Analysis.

Spring Semester – Visva-Bharati University

January 2022 – July 2022

- BSC-SEM-6 CC-4: Algebra.
- BSC-SEM-6 CC-14: Statistical Computing.
- MSC-23: Regression Analysis.

Fall Semester – Visva-Bharati University

August 2022 – December 2022

- MSC-21: Measure Theory and Advanced Real Analysis (M.Sc.).

- BSC-SEM-6 CC-14: Real Analysis (B.Sc.).
- BSC-SEM-6 CC-4: Algebra (B.Sc.).

Spring Semester – Visva-Bharati University

January 2023 – July 2023

- Ph.D. Coursework: Bayesian Inference (Doctoral Level Teaching).
- MSC-21: Regression Techniques (M.Sc.).
- BSC-SEM-6 CC-14: Statistical Computing with C and C++.
- BSC-SEM-6 CC-4: Algebra.

Fall Semester – Visva-Bharati University

August 2023 – December 2023

- BSC-GEC 4: Statistical Computing.
- BSC: Sampling Distribution.
- MSC-SEM-23: Measure Theory and Advanced Real Analysis.

Spring Semester – Visva-Bharati University

January 2024 – July 2024

- BSC-SEM-6 CC-14: Real Analysis.
- BSC-GEC 4: General Elective (SQC, OR, Demography).
- MSC-SEM-23: Regression Techniques.

Fall Semester – Visva-Bharati University

August 2024 – December 2024

- BSC-GEC 4: General Elective (SQC, OR, Demography).
- MSC-SEM-23: Measure Theory and Advanced Real Analysis.
- BSC-GEC 7: Statistical Computing (Old Syllabus).
- BSC-GEC 3: Statistical Computing (New Syllabus).

Spring Semester – Visva-Bharati University

January 2025 – July 2025

- BSC-SEM-6 CC-14: Real Analysis.
- BSC-AECC: General Elective (Probability).
- MSC-SEM-23: Regression Techniques.
- Workshop: Statistical Machine Learning and Artificial Intelligence.

Fall Semester – Visva-Bharati University

August 2025 – December 2025

- BSC-SEM-6: Statistical Computing with C++.
- BSC-AECC: General Elective.
- MSC-SEM-2: Measure Theory and Advanced Analysis.
- MSC-SEM-4: Advanced Data Analysis Techniques.

Spring Semester – Visva-Bharati University

January 2026 – June 2026

- Ph.D. Coursework: Bayesian Inference (Doctoral Coursework Teaching).
- MSC-SEM-23: Regression Techniques (M.Sc.).
- BSC-SEM-5: Statistical Quality Control (SQC).
- BSC-SEM-3: Time Series.

INVITED LECTURE /CONTRIBUTED PRESENTATIONS IN SEMINARS/WORKSHOPS

13. **Invited Lecture:** Debashis Chatterjee. 2026. Invited Lecture at the International Seminar on **Modelling Dynamic Processes under the Confluence of Statistics, Data Science and Machine Learning**. Lecture titled **AI- and Machine LearningDriven Data Science Approaches for Modelling Dynamic Systems**.
12. **Invited Lecture:** Debashis Chatterjee. 2024. An Invited lecture on using statistics to understand geological problems - on May 2025, The Geological Studies Unit Indian Statistical Institute, Kolkata.
11. **Invited Lecture Series on Nonparametric Hypothesis Testing and Statistical Machine Learning**, organized by the Department of Agricultural Statistics, Visva-Bharati University, Santiniketan. Delivered as part of a National Workshop on Advanced Statistical Methods and Applications. Scheduled on **19 February 2025** and **26 February 2025**.
Topics covered: Rank-based Hypothesis Testing, Kernel Methods, Supervised and Unsupervised Machine Learning Techniques.
10. Debashis Chatterjee. 2024. Invited Lecture in **Calcutta Statistical Association Bulletin**. Eleventh International Triennial Calcutta Symposium on Probability and Statistics, organized jointly by the Department of Statistics, University of Calcutta, and Calcutta Statistical Association, December 27 – 30, 2024.
9. Debashis Chatterjee. 2024. An Invited lecture on using statistics to understand geological problems - on 29th May 2024 and demonstrate hands-on the usage of statistics in deciphering the paleoclimatic condition for the short course (on the quantitative aspect of Geology) as an invited speaker of the workshop on Quantitative Palaeontology conducted by The Geological Studies Unit Indian Statistical Institute, Kolkata.
8. Debashis Chatterjee. 2023. An Invited lecture on Landmark and Shape Analysis in Natural Science Data on 22 August 2023, at the Seminar Room of the Human Genetics Unit (Kolmogorov Building), as an invited speaker of the workshop on Quantitative Palaeontology conducted by The Geological Studies Unit Indian Statistical Institute, Kolkata, from 21 to 24 August 2023.
7. Invited Tutor Instructor: (2022 July) Invited as Tutor Instructor for Tutorial Classes on Unsupervised Learning at Instructional School for Teachers at Indian Statistical Institute
6. Debashis Chatterjee. 2020. An Invited Lecture on Bayesian Inverse Regression application on Geology & paleontology. Organized Seminar: Fossils of Bengal: by Geology Studies Unit, Indian Statistical Institute, Kolkata, India.
5. Debashis Chatterjee, 2020 (postponed due to COVID-19) On Bayesian Inverse Regression at Conference of ISIISMISSAS Joint Statistical Meeting 2020. Oral presentation, Bayesian Inverse Regression, Poster: Event Presented At Indian Statistical Institute, Kolkata, India.
4. Debashis Chatterjee. 2019. National Seminar on Application of Statistics on Natural Science. (IUCAA Center for Astronomy Research & Development), St Xaviers College (Autonomous), Kolkata
3. Debashis Chatterjee. 2018. Presentation on Bayesian Inverse Regression at CALCUTTA UNIVERSITY (10th ITCS 2018)
2. Debashis Chatterjee. 2017. On Bayesian Inverse Regression. Departmental seminar: Indian Statistical Institute
1. National and International Statistical Systems: joint group project on data analysis and interpretation of data provided by NSSO.(National Sample Survey Office).

ACADEMIC LEADERSHIP, CONFERENCE ORGANISATION & INSTITUTIONAL SERVICE

5. **Joint Convener (2025): National Seminar on Recent Advances in Statistics: Theory and Applications**, Department of Statistics, Visva-Bharati University, Santiniketan (Online, 1718 March 2025). Coordinated academic planning, technical management, and overall execution.
4. **Acting Head of the Department (Temporary Charge):** Department of Statistics, Visva-Bharati University. Undertook administrative and academic responsibilities during the absence of the designated HoD, including official meetings and departmental coordination.
3. **Member, Green Committee (Visva-Bharati University):** Contributing to institutional sustainability initiatives, environmental planning, and campus-level awareness programmes.
2. **Academic/Administrative Coordination:** Library-in-Charge, Supervision and coordination of M.Sc. and B.Sc. dissertation projects; liaison with examination bodies and evaluation committees.
1. **Departmental Committees:** Organizing Committee Member; Board of Studies Committee Member; Library-in-Charge; coordination of departmental academic and outreach activities.

Research Proposals Submitted / Institutionally Endorsed / Under Consideration

The following proposals are listed as submitted, institutionally endorsed, or under consideration. They are not claimed as sanctioned/funded projects unless an official sanction order is received.

S.No	Scheme / Agency	Title of Proposal	Role	Collaborators / Team	Status
1	ANRF AI for Science & Engineering / AI Weather and Climate Modelling	Spherical & Bayesian Inverse Intelligence for Indian Weather, Climate, and Geological Hazards	Lead PI	Prof. Amlan Banerjee, Geological Studies Unit, Indian Statistical Institute, Kolkata	Submitted / under consideration
2	ANRF ARG-MATRICES	Stochastic Directional Operations Research on Manifolds: Distributional Theory, Bayesian Extension, Geodesic Risk and Applications in Astronomy and Bioscience	PI	—	Submitted under ANRF ARG-MATRICES
3	ANRF Advanced Research Grant / ARG	Higher-Order Measure-Theoretic Differentiation and Singularity Geometry: A Unified Calculus for Irregular Scientific Data, Geometric Measures and India-Relevant Extreme Phenomena	Co-PI	PI: Dr. Subhasis Roy	Departmentally recommended / forwarded for institutional endorsement
4	Visva-Bharati Internally Funded Research Grant for Young Faculties, 2026	AI-Mediated Interpretive Responses to Tagore and Wordsworth across Transcendental, Relational and Environmental Axes	Co-PI	PI: Dr. Surhita Basu; Co-PIs: Dr. Shinjini Basu and Dr. Debashis Chatterjee	Submitted / processed internally at Visva-Bharati

Participation in UGC Orientation / Refresher Courses

3. **Refresher Course: Artificial Intelligence for Education & Research**
Malaviya Mission Teacher Training Centre, The University of Burdwan (03 November – 18 November 2025).
Focused on AI-driven pedagogical innovation and research integration.
2. **Refresher Course: Emerging Dimensions in Biological Sciences**
UGCHRD, University of North Bengal (24 November – 08 December 2023).
Advanced exposure to interdisciplinary biological research methodologies and academic development.
1. **9th Faculty Induction Programme**

Scholarly and Creative Interests

4. **Fine Arts:** Practising visual artist (drawing and painting); participant in curated exhibitions.
3. **Classical Music:** Formal training in Classical Music & Rabindrasangeet; instrumental knowledge in violin, tabla, and basics of guitar; currently pursuing advanced learning in Esraj, violin, and bansuri.
2. **Literature:** Bengali fiction writer with published works and ongoing manuscripts/ story books with reputed publishers and literary platforms.
1. **Languages:** English, Bengali, Hindi, Sanskrit; foundational knowledge of Classical Greek and Latin.



Dr. Debashis Chatterjee
Assistant Professor, Department of Statistics
Visva-Bharati University, Santiniketan